GeoDeducr – Functional Design and Mock-up

Mobile Applications for iOS · Module 1 · Oct 26, 2026 · German Moll 288388

Application overview

GeoDeducr is an iOS geography game in which the player finds a hidden city on a world map by reasoning from facts about it, not from photos of it. Each round reveals clues one at a time, such as what the city is famous for, its local dish, the language spoken there, or its current weather and local time, and the player drops a pin where they think the city is. The closer the pin, the higher the score.

Target users: students, travellers and geography quiz fans who enjoy GeoGuessr-style games but want to rely on knowledge and deduction rather than visual recognition.

Goals of the application:

- Turn geography facts into a short, replayable deduction game (rounds of about 1 minute).
- Combine permanent facts with live data, so the same city plays differently on different days.
- Reward reasoning: fewer clues used means more points.

Project type: individual project. The mock-up has 8 screens (13 Figma frames including onboarding pages, states and overlays), above the 6-screen mock-up minimum and the 3-screen minimum for one person.

Gameplay

A game has 5 rounds. Every target is a city, never a whole country. Each round hides one city and gives the player up to 5 clues, from vague to specific. A round works like this:

1. The first clue is revealed automatically.
2. The player can drop a pin on the map at any time. In untimed modes (Classic, Few hints) the player unlocks the next clue by tapping it. In timed modes (Speed run, Blitz, Daily challenge) a new clue arrives automatically every few seconds, and the player can pay points to unlock it early.
3. Each unlocked clue lowers the maximum points for the round.
4. After the pin is confirmed, the app shows the real location, the distance and the points earned.

Clue types

Each city has a pool of facts, and every round draws a random mix from it, so the same city gives different clues each time. Clues come in a fixed order, from the vaguest to the most revealing.

Step	Clue type	Example (Paris)	Media
1	Live fact	"It's 14°C and cloudy, local time 15:40"	Animated weather icon
2	Geography	About 2.1 million people, a river runs through it	Text card

Step	Clue type	Example (Paris)	Media
3	Language and culture	"Bonjour" spoken aloud; pays in euro	Audio clip
4	Local food	Croissants and steak frites	Image card
5	Famous for	Home to a wrought-iron tower built for the 1889 World's Fair	Illustration card

"Famous for" clues are the city's best-known facts: a landmark, a festival, a sports club, a famous person or a nickname. They nearly give the answer away, so they always come last. Live clues come first because they narrow the region without naming anything: "it is 04:12 and still dark" points to a longitude, and "it is summer in October" points south of the equator.

Scoring: up to 5,000 points per round. A pin within 25 km of the city scores full points, and points fall as the distance grows. Each extra clue unlocked lowers the maximum (5,000 → 4,000 → 3,000 and so on). In timed modes, the pin on the map when time runs out counts as the guess; no pin scores 0.

After round 5 the game total earns 1 to 3 stars: 1 star from 5,000 points, 2 from 12,000 and 3 from 20,000.

What the HUD shows: the star pill in the top-right corner of the Game screen is the most the current round can still earn, not a balance to spend. It starts at 5,000 and drops by 1,000 each time a clue is unlocked, so the "−1,000" on a locked clue card shows how much the maximum will fall. When the guess is confirmed, the distance penalty is taken from this maximum to give the round score (Paris at 412 km with 3 clues: 3,000 − 150 = 2,850). The trophy marks totals: the game total during a game, and the all-time Total Score on Home.

Game modes

All modes are city-based. They differ only in how many clues the player gets and how much time they have.

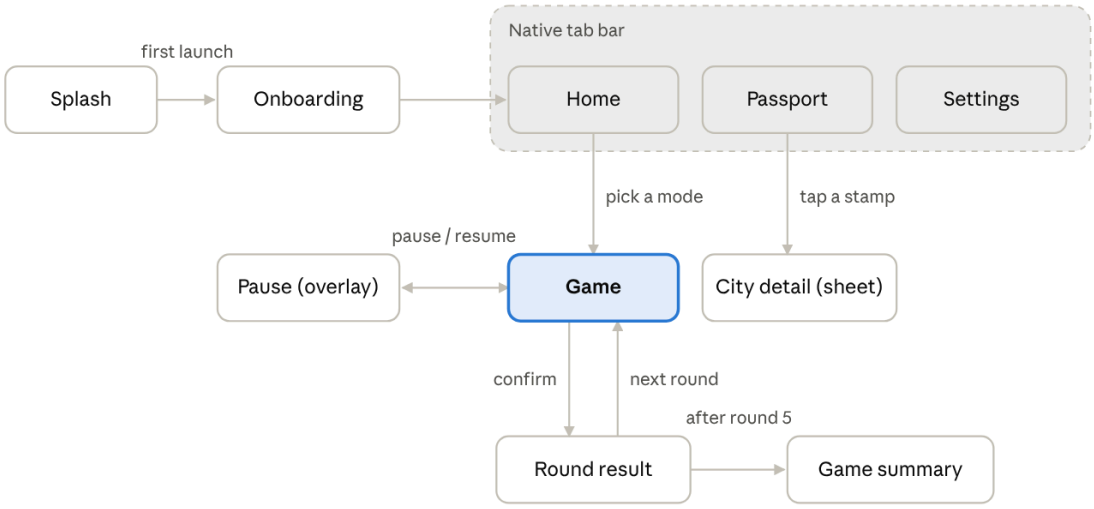
Mode	Clues per round	Time per round	Notes
Classic	5	No limit	The standard game
Few hints	3	No limit	Steps 4 and 5 (food, famous for) are never shown
Speed run	5	30 s	All clues available, but little time to read them
Blitz	2 (random)	15 s	Hardest mode
Daily challenge	5	60 s	The same 5 cities for every player, once a day

Screens

The app has 8 screens. Home, Passport and Settings are the three tabs of a native iOS tab bar; a game runs full-screen on top of them, looping Game → Round result five times before the summary.

#	Screen	Purpose	Main elements
1	Splash	Shown for about 1 second at launch	Logo with spinning globe, bouncing loading dots
2	Onboarding	Teaches the game on first launch (replayable from Settings)	Page 1: tutorial video with Skip; pages 2–4: rule cards (read the clues, drop a pin, closer = more points)
3	Home (tab)	Choose how to play	Two-tone logo, wooden board with 5 mode cards, total score; after today's Daily Challenge its card shows the score and a countdown
4	Game	Read clues and guess	Full-screen map limited to country-level zoom, pause button (top-left), timer and points (top-right), clue sheet with the newest clue and a countdown bar, expands to list all clues, "Confirm guess" button. Overlay: pause card over a blurred map (Resume, Restart round, Sound, Quit to Home)
5	Round result	Show how close the guess was	Map zooms out, dashed line from the pin to the city, result card with city photo, distance and points breakdown (max points for clues used, distance penalty, round score), "Next round"
6	Game summary	Results of the 5 rounds	World map with all guesses and cities, total score with stars, list of rounds, Play again, Home and Share (emoji grid via the iOS share sheet)
7	Passport (tab)	Collection of cities guessed	Stats (cities, games, best score), Stamps/List switch, stamp grid by continent with "?" for locked cities. Overlay: city detail sheet with photo, fun fact, best score and a "Hear hello" audio button
8	Settings (tab)	Preferences	Native grouped list: sound effects and music toggles, distance and temperature units, replay tutorial, reset progress (with confirmation)

Three tabs; each game loops Game and Round result five times



navigation flow · 8 screens

Later launches skip Onboarding (Settings can replay it). The summary returns to Home or starts a new game with Play again.

Screen transitions

Every action that moves between screens or changes a screen's state, with the animation planned for it.

From	Action	To	Animation
Splash	After about 1 second	Onboarding (first launch) or Home	Fade
Onboarding	Next or swipe	Next page; the last page's "Let's play!" opens Home	Page slide
Onboarding	Skip	Home	Fade
Home, Passport, Settings	Tap a tab	The other tab	Native tab switch
Home	Tap a mode card	Game	Full-screen cover slides up
Game	Tap the map	Pin placed, "Confirm guess" appears	Pin drop, button pops in
Game	Drag the clue sheet up or down	Clues expanded or collapsed	Sheet spring

From	Action	To	Animation
Game	Countdown ends or paid unlock	Next clue shown	Card flip
Game	Tap pause	Pause	Map blurs, card fades in
Pause	Resume or Restart round	Game	Blur fades out
Pause	Quit to Home, then confirm	Home	Alert, then cover closes
Game	Confirm guess, or time runs out	Round result	Map zooms out, line draws, sheet becomes the result card
Round result	Next round (rounds 1–4)	Game	Map resets, sheet returns to clues
Round result	See results (round 5)	Game summary	Slide
Game summary	Play again	Game (new game)	Cross-fade
Game summary	Home	Home	Cover closes
Game summary	Share	iOS share sheet	Native sheet
Passport	Stamps / List switch	The other view	Cross-fade
Passport	Tap a stamp or row	City detail	Sheet slides up
City detail	Close or swipe down	Passport	Sheet slides down
City detail	Hear "Bonjour"	Audio plays	Speaker icon wiggles
Settings	Replay tutorial	Onboarding	Full-screen cover
Settings	Reset progress, then confirm	Scores and stamps cleared	Confirmation dialog

Multimedia requirements

Every course requirement is covered by a part of the game that needs it anyway.

Requirement	How GeoDeducr meets it	Screen
At least 15 images	About 100 city photos, a stamp per city, 20+ illustrated icons and the logo	Round result, City detail, Passport, Home

Requirement	How GeoDeducr meets it	Screen
At least 1 video	Tutorial video (tutorial.mp4) played inline with AVKit	Onboarding (replayable from Settings)
At least 1 audio file	"Hello" spoken in the local language (e.g. bonjour.m4a) as a clue and in City detail; sound effects for pin drop, clue reveal and results; background music	Game, City detail, Round result
Animations	Spinning globe, clue cards flipping in, pin drop, distance line drawing, score counting up, timer pulsing in the last 10 seconds	Splash, Game, Round result, Game summary
Runs on an iOS device	SwiftUI app for iPhone, tested on a physical device	All

All images and audio will be royalty-free or self-made, and bundled in the app.

Visual design

GeoDeducr looks like a friendly toy: every element has a thick dark-brown outline, rounded corners and a solid "lip" underneath, so buttons look pressable. Everything is built from shared Figma variables, text styles and components, and each maps directly to SwiftUI.

Colour palette

Role	Colour	Used for
Sky	#5EB8F2 → #E6F8FF	Screen backgrounds
Lawn	#4FA83D	Hills at the bottom of Home, Splash and Summary
Cream card	#FFF6E5	Cards, sheets, panels
Wooden board	#A0581F	Mode board, locked clue card, stamp codes
Outline and main text	#3B1F0A	All outlines and body text
Secondary text	#6B3A12	Captions, labels
Accent green	#2E7D32	Primary buttons, toggles, selected tab, progress bar
Logo "Geo" / "Deducr"	#7ED957 / #D9A066	Two-tone wordmark
Danger	#B91C1C	Quit, reset progress, low timer, penalties
Mode colours	Gold #FFC83D, green #2E7D32, orange	Daily, Classic, Speed run, Few hints, Blitz

Role	Colour	Used for
	#C2410C, blue #1E5BC6, red #B91C1C	

Typography: Luckiest Guy for the logo only; Nunito (SemiBold to Black) for all other text, on the iOS Dynamic Type sizes from Large Title (34 pt) to Caption (12 pt), so text scales with the user's settings.

Components: custom game components (Mode Card, Clue Card, Chunky Button, Icon Button, HUD Pill, Timer Pill, Clue Progress, Score Badge, Stat Tile, City Stamp, Map Pin and 20 illustrated icons) plus Apple's iOS 27 components for system UI: status bar, tab bar, toggles, segmented picker, grouped lists and sheets, all tinted with the accent green.

Accessibility: white text sits only on the darker mode colours and dark-brown text on gold, all at a contrast of at least 4.5:1; tap targets are at least 44 pt; city photos and icons get VoiceOver labels.

Technical outline

GeoDeducr needs no backend or user accounts. Almost everything runs locally, and only one feature uses the network.

- **UI:** SwiftUI with Apple's native components: TabView (Home, Passport, Settings), sheets with two heights for the clue panel, Form for Settings and a segmented Picker in Passport. One `.tint(Color.accent)` colours every system control; the app runs in light mode.
- **Map and guessing:** MapKit with the camera zoom limited to country level, so city names never appear; tap to drop the pin; `CLLocation.distance(from:)` for the distance.
- **City data:** a bundled JSON file with about 100 cities (coordinates, clue pool, fun fact, photo and audio file names).
- **Live clues:** current weather from the free Open-Meteo API, requested by the city's coordinates; local time and day or night computed from the time zone. If there is no internet, the app falls back to permanent clues only.
- **Media:** AVKit for the tutorial video, AVAudioPlayer for the "hello" clips and sound effects.
- **Sharing:** ShareLink opens the iOS share sheet with the result as an emoji grid.
- **Saved progress:** scores, Passport stamps and settings stored locally on the device.

Lab requirements check

- Mock-up of at least 6 screens: 8 screens, 13 Figma frames.
- One-person project, at least 3 screens: 8 screens.
- At least 15 images, 1 video, 1 audio file and animations: see Multimedia requirements.
- Runs on an iOS device: native SwiftUI iPhone app.
- Tools: Figma for the mock-up; this document is exported to PDF.
- Submission: `IOS_<indexNumber>_GeoDeducr_2026.zip` sent via Microsoft Teams.

Mock-up presentation

The full mock-up is in Figma: GeoDeducr mock-up, section "GeoDeducr · Mock-up (Module 1)". Scores, stats and city counts on the screens are sample data (for example, Passport shows 18 of 50 cities while the app will ship with 100), and city photos are placeholders for royalty-free photos.

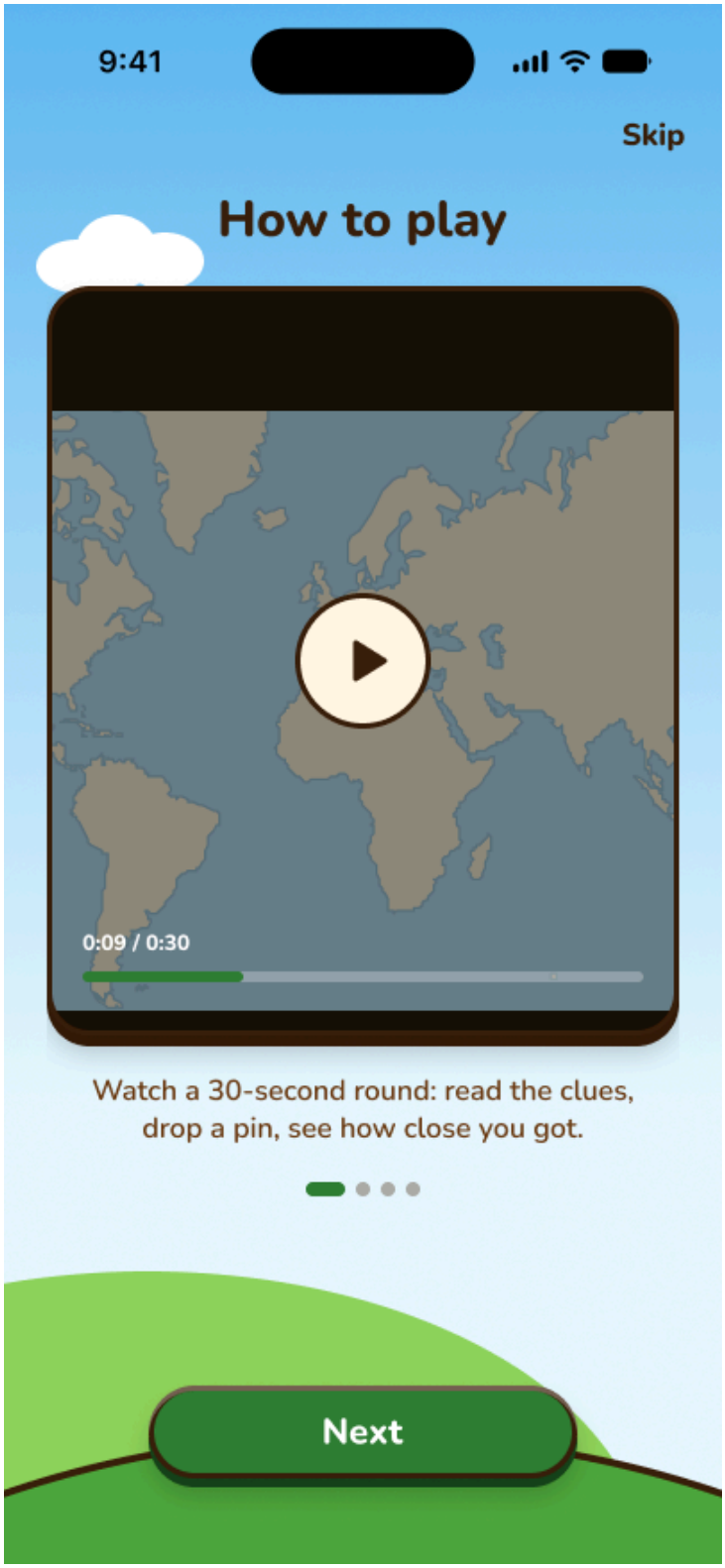
1. Splash

Logo with a spinning globe while the app loads, about 1 second.



2. Onboarding – Video

First launch: a 30-second tutorial video plays inline, the required video of the project.



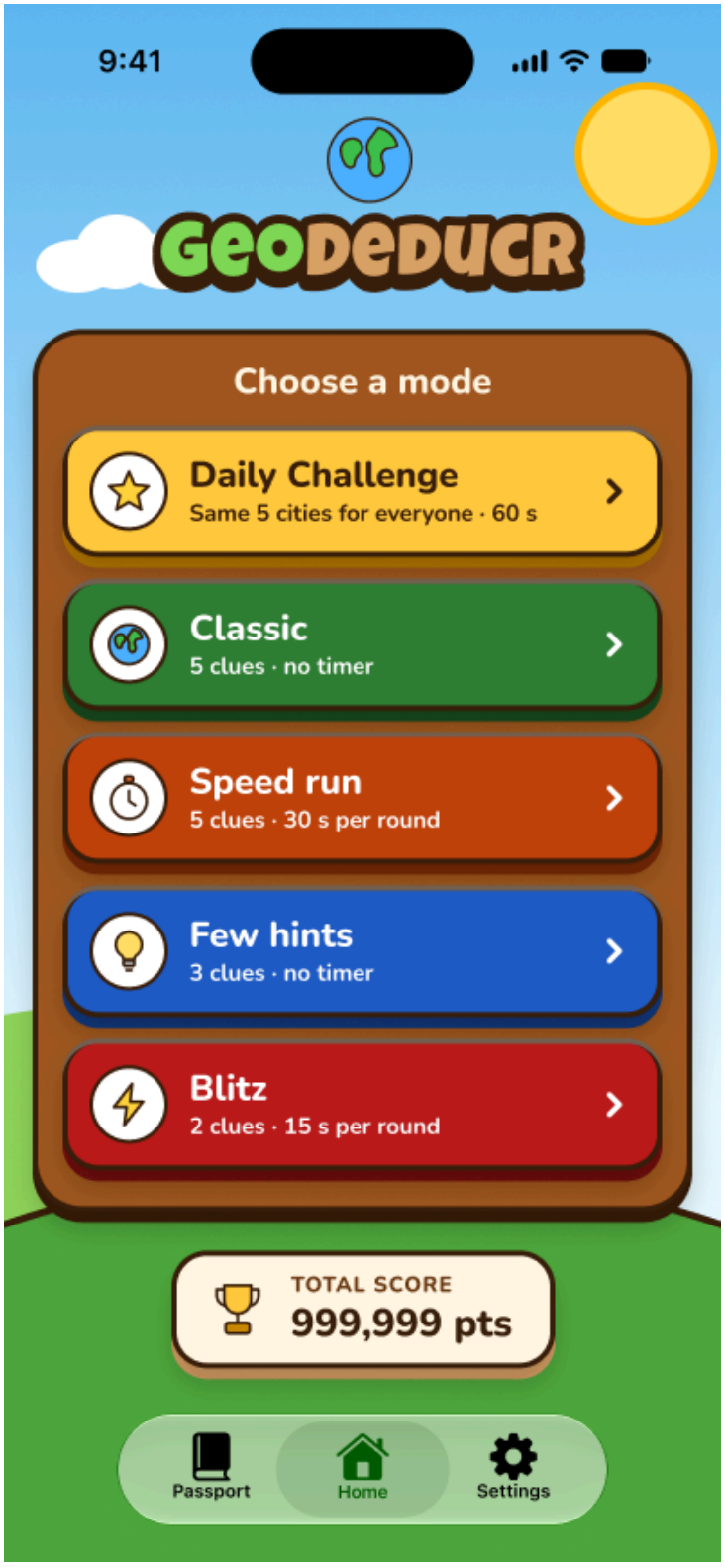
3. Onboarding – Rules

Three rule cards: read the clues, drop a pin, closer means more points.



4. Home

The five game modes on a wooden board, with the total score and the native tab bar.


9:41
GeoDeducr
Choose a mode
Daily Challenge
Same 5 cities for everyone · 60 s
Classic
5 clues · no timer
Speed run
5 clues · 30 s per round
Few hints
3 clues · no timer
Blitz
2 clues · 15 s per round
TOTAL SCORE
999,999 pts
Passport
Home
Settings

5. Game – clues collapsed

Map-first view: pause top-left, timer and points top-right, newest clue in the sheet, "Confirm guess" after placing the pin.


9:41
0:24
3,000
DENMARK
IRELAND
UNITED KINGDOM
NETHERLANDS
POL
GERMANY
BELGIUM
CZECHIA
AUSTRIA
FRANCE
SWITZERLAND
ITALY
PORTUG
Confirm guess
ROUND 2 OF 5
Next clue in 0:08
CLUE 3 · CULTURE
People greet you with "Bonjour" and pay in euro.

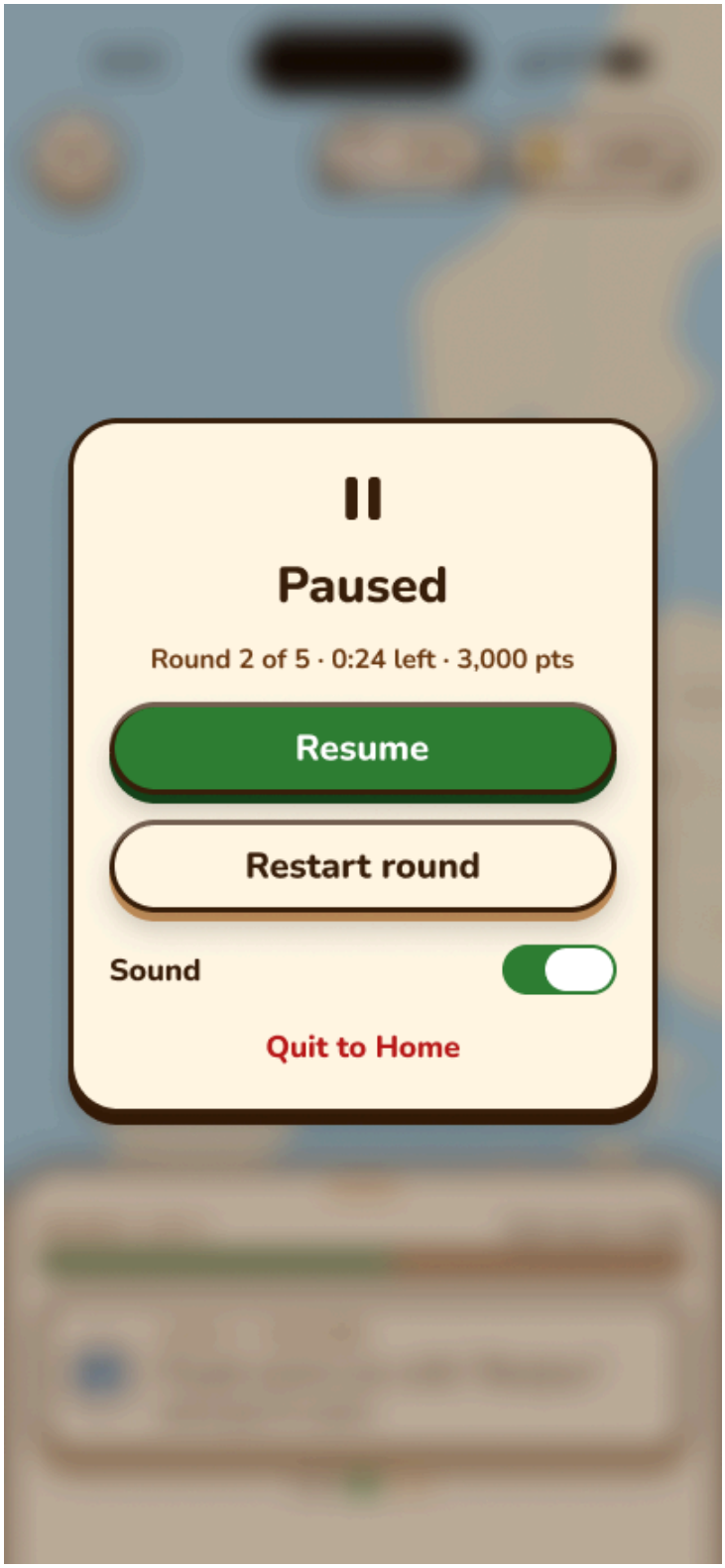
6. Game – clues expanded

The sheet pulled up shows every clue so far and the countdown to the next one, which can be unlocked early for points.



7. Pause

The map blurs and the timer stops: Resume, Restart round, Sound and Quit to Home.



8. Round result

A line from the guess to the real city, the city photo and the points breakdown.



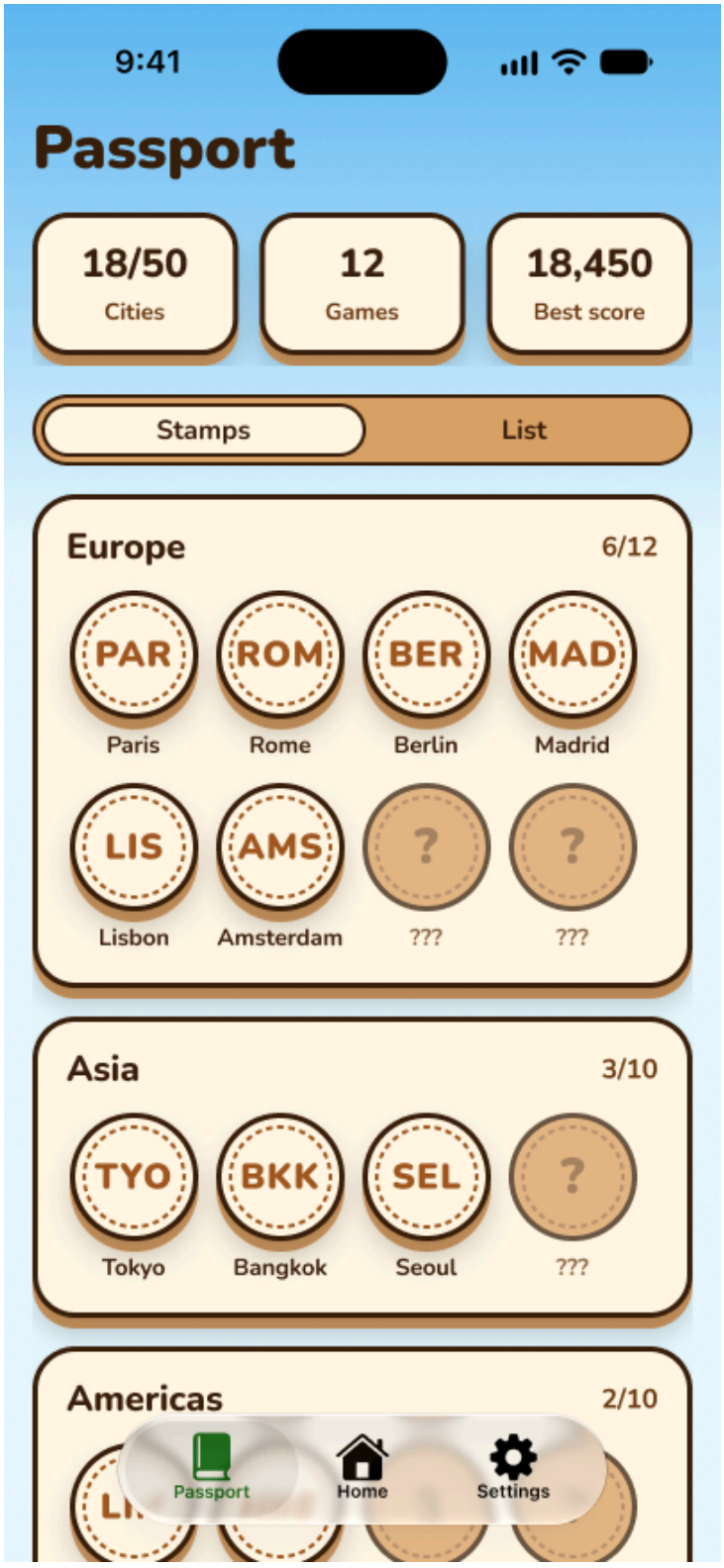
9. Game summary

All five rounds on a world map, the total score with stars, and Play again, Home and Share.



10. Passport – Stamps

Stats, then a stamp for every city collected, grouped by continent; "?" marks cities still to find.



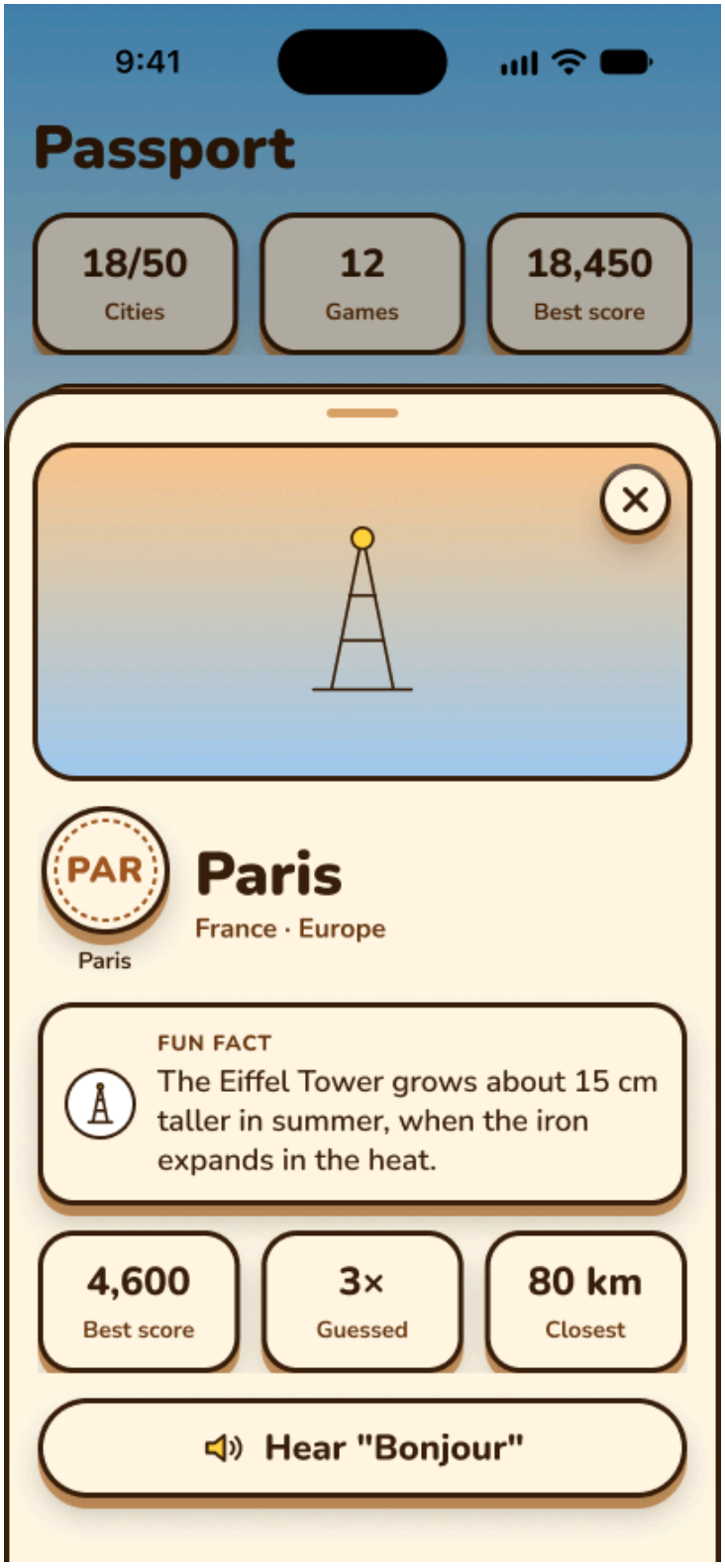
11. Passport – List

The same collection as a native list with the best score per city.



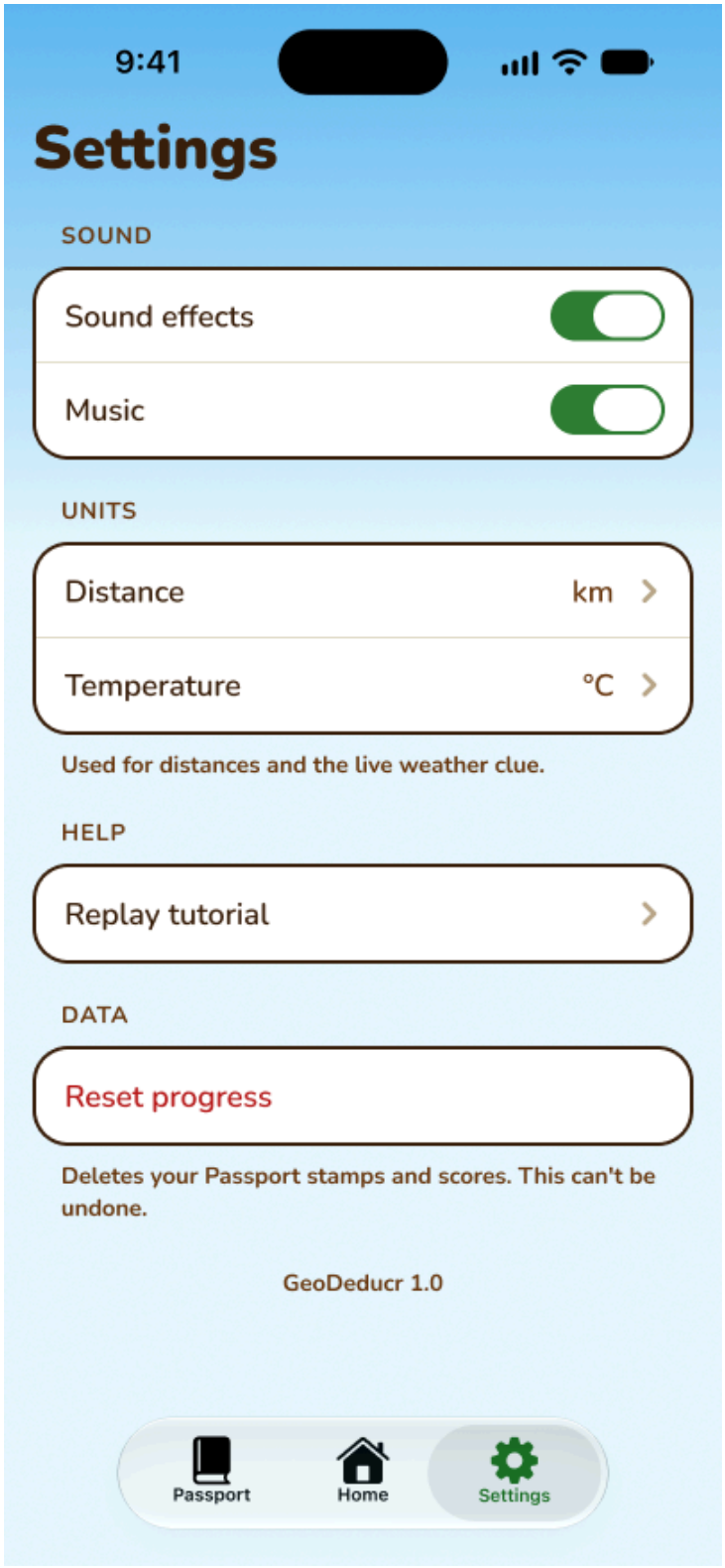
12. City detail

City photo, fun fact, personal stats and the "Hear Bonjour" audio button.



13. Settings

Sound, units, replay tutorial and reset progress in a native grouped list.


9:41
Settings
SOUND
Sound effects
Music
UNITS
Distance
km
Temperature
°C
Used for distances and the live weather clue.
HELP
Replay tutorial
DATA
Reset progress
Deletes your Passport stamps and scores. This can't be undone.
GeoDeducr 1.0
Passport
Home
Settings